Vision AI for Defense

From camera views to useful insights

Use Isaac Sim to create realistic environments with drone, fixed and vehicle-mounted cameras. Bring the camera views into VSS to show what edge AI can help an operator see, understand and review.

01 Drone inspection and search

- Use Pegasus in Isaac Sim to create camera-equipped drones that fly through an airfield or facility.
- Find people, loose objects or surface issues. Select an object for the AI to describe or outline with segmentation.
- In VSS, move from a survey view to a closer image, then collect the findings in an inspection report.

02 Site security

- Place cameras around an air base, including hangars, gates and storage areas.
- Detect and track people across camera views. Define areas that trigger an alert when someone enters, crosses a boundary or stays too long.
- In VSS, open the alert, follow the event sequence and review the supporting images or video.

03 Vehicle awareness

- Attach front, rear and side cameras to a vehicle in Isaac Sim.
- Detect people, vehicles and obstacles around it. Bring the relevant camera view forward when something needs attention.
- In VSS, show the surrounding camera views together, with visual cues that help the driver understand where to look.

04 Aircraft exterior inspection

- Fly a camera around an aircraft and highlight visible panel, seal or coating issues for closer review.
- In VSS, browse the inspection images and open each highlighted area for the maintenance team.

05 Border security

- Use fixed or drone cameras to monitor remote border areas and access roads. Detect people and vehicles, track movement and flag crossings of a defined boundary.
- In VSS, bring the relevant camera view and alert forward, then show the event sequence and supporting images for operator review.

How this fits into VSS

VSS — Video Search and Summarization — brings the camera views and AI results into one operator workspace. Watch the views, inspect findings and ask questions with supporting images or clips.